

Descriptives: combined_9_17_26

The full medical-device 10-K/10-Q panel (one row per filing) with FDA product events and event 8-Ks merged in by gvkey $\times$ reporting year $\times$ reporting quarter.

Source: `data/combined_9_17_26.csv`.

Built by `scripts/18_build_combined_filing_event_dataset.py`.

Overview

Statistic	Value
Filings = firm reporting quarters (rows)	6,547
10-K	1,660
10-Q	4,887
Firms (gvkey)	117
Reporting periods ending	2004-11-27 to 2026-05-31
Filings with a parsed MD&A section	6,238
Filings with a warning letter in the quarter	92
Filings with a recall in the quarter	1,376
Filings with an adverse-event report in the quarter	4,059
Filings with no product event of any type in the quarter	2,367
Filings with an event 8-K in the quarter	74

Unique filings with a SUBSTANTIVE mention of each event in the MD&A

Event substantively discussed in MD&A	10-K filings	10-Q filings	All filings	Share of all filings
Warning letter	135	461	596	9.1%
Recall (Part 806)	272	581	853	13.0%
Adverse event (Part 803)	108	166	274	4.2%
Any of the three (each filing once)	367	891	1,258	19.2%

Counts are unique filings (accession numbers) across the FULL baseline panel, MD&A section only (Item 7 in a 10-K, Item 2 in a 10-Q). 'Substantive' = at least one MD&A occurrence tied to an actual event (past-tense receipt/action verb, definite reference, or specific date) and not hedged by a modal - script 17's definition. A filing can count under more than one event type.

Substantive MD&A disclosure: quarters WITH vs WITHOUT the event

Event type	Filings: event in quarter	... substantive MD&A	Rate	Filings: no event in quarter	... substantive MD&A	Rate
Warning letter *	91	34	37.4%	5,117	473	9.2%
Recall (Part 806)	1,376	291	21.1%	5,171	562	10.9%
Adverse event (Part 803)	4,059	199	4.9%	2,488	75	3.0%

*Raw rates, no controls - descriptive only. 'Event in quarter' = at least one event of that type dated inside the filing's own reporting period. Discussion in a no-event quarter is mostly the continuing discussion of an EARLIER quarter's event, so the right-hand rate is not a clean counterfactual. * Warning-letter row uses only filings whose whole period falls on/after 2008-10-01, when FDA's letter data begins; earlier filings have letters UNOBSERVED, not absent, and are excluded rather than counted as no-event.*

Substantive MD&A disclosure at the YEARLY (10-K) level: fiscal years WITH vs WITHOUT the event

Event type	10-Ks: event in fiscal year	... sub- stantive MD&A	Rate	10-Ks: no event in fiscal year	... sub- stantive MD&A	Rate
Warning letter *	86	21	24.4%	1,184	93	7.9%
Recall (Part 806)	626	144	23.0%	1,034	128	12.4%
Adverse event (Part 803)	1,166	90	7.7%	494	18	3.6%

10-K filings only. 'Event in fiscal year' = at least one event of that type dated anywhere inside the fiscal year the 10-K covers (previous 10-K's period end to this one's; all four quarters, not just Q4). Raw rates, no controls. * Warning-letter row uses only filings whose whole period falls on/after 2008-10-01, when FDA's letter data begins; earlier filings have letters UNOBSERVED, not absent, and are excluded rather than counted as no-event.

Event 8-Ks by event type

8-K event type	Documents (forms + exhibits)	Unique 8-Ks	... at baseline- panel firms	... merged to a reporting quarter	Filings (quarters) with such an 8-K
Warning letter	151	115	65	62	60
Recall (Part 806)	72	62	16	15	15
Adverse event (Part 803)	7	6	5	5	5
Any event (each 8-K once)	222	176	80	76	74

Source: the device 8-K dataset (script 16), classified on the filing lead. One 8-K can carry several documents (form + exhibits) and more than one event flag. Only 8-Ks filed by firms in the 10-K/10-Q baseline panel can merge; an 8-K is placed by its reported event date.

Product event dates (the events themselves - no filings involved)

Event type - scope	Events	Distinct firm-dates	Firms	First event date	Last event date
Warning letter - all gvkey-linked firms	192	188	135	2008-10-10	2026-04-30
Warning letter - firms in the 10-K/10-Q panel	128	125	86	2008-10-10	2026-04-30
Warning letter - merged to a reporting quarter	96	93	67	2008-10-21	2026-01-29
Recall (Part 806) - all gvkey-linked firms	17,448	5,411	113	2000-04-11	2026-06-30
Recall (Part 806) - firms in the 10-K/10-Q panel	11,722	3,989	82	2000-04-11	2026-06-30
Recall (Part 806) - merged to a reporting quarter	7,761	2,544	74	2004-09-20	2026-03-30
Adverse event (Part 803) - all gvkey-linked firms	10,737,528	266,574	177	1991-12-31	2026-06-30
Adverse event (Part 803) - firms in the 10-K/10-Q panel	9,588,496	202,984	113	1991-12-31	2026-06-30
Adverse event (Part 803) - merged to a reporting quarter	7,114,479	118,510	110	2004-09-03	2026-05-28

Counted from the FDA source data. Event = one warning letter (unique Case ID, dated by action date), one recall record (dated when the firm initiated it), one adverse-event report (dated when FDA received it). 'Distinct firm-dates' counts each firm x calendar date once. Events not merged belong to firms outside the 10-K/10-Q panel, predate the firm's first filing in it (the panel starts in 2005; MAUDE in 1991), postdate its last, or fall in a gap between filings.

Industry coverage: share of the US publicly traded medical-device industry (Table 1 of 2026-07-06, carried over - NOT recomputed)

Statistic	Value
Number of warning letters	120
Number of unique firms	81
Firm market capitalization (\$M), mean	6,649
Firm market capitalization (\$M), median	1,329
Firm total assets (\$M), mean	5,942
Firm total assets (\$M), median	439
Share of US medical-device market capitalization	60.3%
Sample period	October 2008 – January 2026

CARRIED OVER VERBATIM from output/tables/table1_device_sample_descriptives_2026-07-06 and not recomputed for this dataset. It describes the linked device WARNING-LETTER sample (120 letters, 81 firms), not the 117-firm 10-K/10-Q panel above. Original notes: This table describes the analysis sample: FDA warning letters classified by the FDA as medical-device actions whose recipient is linked to a Compustat firm (gokey) with an active record as of the letter date — a non-missing market capitalization at the last fiscal year-end before the letter (within 18 months). Links come from the project crosswalk, inclusive of matches flagged for manual review; ownership windows are respected, so a letter links to the firm that owned the recipient in the letter year. Letters are counted by unique FDA Case/Injunction ID. Market capitalization (fiscal year-end close price times common shares outstanding) and total assets are measured at the last fiscal year-end before the firm's first letter, in \$ millions. The market-cap share is the sample firms' current market capitalization as a fraction of the US medical-device universe: 26 sample firms of the 182 US-incorporated Compustat firms with primary SIC 3841–3845 and a market cap at their latest fiscal year-end; sample firms delisted or acquired since their letter no longer contribute. Sources: FDA Data Dashboard compliance actions; Compustat via WRDS.

Caveats

1. Reporting year/quarter is the calendar year and quarter of the period end (snapped to the nearest month end) - a label for the firm's reporting period, not its own fiscal-quarter number. A 10-K row is the fiscal 4th quarter.
2. An event is attached ONLY to the quarter it is dated in. Firms keep discussing a warning letter for years, so lagged/cumulative event indicators are needed before reading the no-event rate as a counterfactual.
3. Adverse-event reports arrive almost every quarter at large firms, so event_adverse_event has little variation there; use the counts (n_adverse_event, and n_mdr_tier_a / n_mdr_tier_b separately - Tier B is parent-prefix name matching).
4. Events use script 14's ownership-window gate but NOT its comp.funda gate; compustat_active_year flags the firm-years that gate would keep.
5. Only Risk Factors and MD&A are parsed (script 17); event discussion located only in Legal Proceedings (Item 3) is missed.